Jacob Cockerill

College Station, TX
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Education:

Master of Science, Mechanical Engineering Expected May 2028
Texas A&M University, College Station, TX
Bachelor of Science, Mechanical Engineering May 2026
Rose-Hulman Institute of Technology, Terre Haute, IN
Concentration in Aerospace Engineering
Certification in the Fundamentals of Engineering: Mechanical

Skills:

Software: Windows OS, Microsoft Office Suite, SolidWorks, ANSYS Mechanical (Static Structural, Modal & Harmonic Response), ANSYS Fluent, MEscope, EES, PicoSoft
Programming: MATLAB, Python, C++

Industry Experience:

Renascent, Inc. (Indianapolis, IN) *Research & Development Engineering Contractor* June 2026 – August 2026

- Developed an inverse surrogate model in Python to estimate payload on CAT 335F and CAT 374F excavators from hydraulic pressure, boom and stick orientation, and motion measurements.
- Used calibration-based regularized regression and motion compensation to model how excavator position and movement affect the pressure-to-payload relationship.
- Implemented signal filtering, outlier rejection, and stability checks to support consistent payload estimates.
- Evaluated model behavior through simulations based on recorded field data: 820 model replays across 164 unique captures and five calibration scenarios, plus 2,750 sensor-disturbance cases generated from 11 representative recordings.
- The R&D team developed an onboard weighing system with a Raspberry Pi touchscreen for bucket weighing and truck-load tracking, sensor integration, installation kits, and field testing on CAT 335F and CAT 374F excavators.

Richmond Sanitary District (Richmond, IN) *Engineer Intern* June 2024 – August 2024

- Developed a spreadsheet system to analyze flow, power, and runtime data across 10 municipal lift stations to establish performance baselines and identify operational inefficiencies using Flowlink Cipher data.
- Verified and corrected as-built drawings against field conditions, updating the district database with unmapped structures.

Winchester Wastewater Treatment (Winchester, IN) *Engineer Intern* June 2023 – August 2023

- Performed routine inspections of treatment-plant and remote-station pumps and logged operating hours and sensor status for maintenance records.

Selected Technical Projects:

Senior Design Project: Excavator Payload Estimation System for Renascent, Inc. September 2025 – May 2026

- Designed and built an excavator payload-estimation prototype using hydraulic pressure transducers, dynamic inclinometers, a Raspberry Pi, and an in-cab touchscreen for approximately $2,500 per excavator.
- Developed a Python calibration and payload-estimation workflow using hydraulic pressure, boom and stick positions, and engine speed during a prescribed lift, without direct geometric measurements.
- Field-tested the prototype across five payload levels and six engine-speed settings to evaluate performance within the prescribed lifting procedure.

Selected Technical Projects Continued:

2D Finite-Volume Solver for Compressible Nozzle Flow April 2026 – May 2026

- Developed a custom 2D finite volume solver in MATLAB to model steady inviscid compressible flow throughout a converging nozzle.
- Implemented a multi-dimensional finite volume scheme, structured skewed-mesh generation, boundary conditions, and a mesh-refinement analysis to resolve the flow-field solution variables.
- Validated the solver against a quasi-1D isentropic nozzle model, achieving maximum relative errors below 1% in Mach number, pressure, density, and temperature while capturing cross-stream flow variation.

Dynamic Modeling & Simulation of a Cincom D25 VIII CNC Automatic Lathe January 2026 – February 2026

- Modeled and simulated the secondary tool-post drive system of a Cincom D25 VIII CNC automatic lathe, including servo motors, belt and chain drives, ball screws, tool positioning, and machining loads.
- Created mathematical models using bond graphs for the X2 and Y2 rotary-tool devices and the dual-motor linear positioning system.
- Derived and parameterized the Y2 drive equations for the electrical, rotational, and linear components, including a compliant tool-to-workpiece interface.
- Simulated step, pulse, and ramp inputs up to 24 V in MATLAB and evaluated motor speed, output force, ball-screw position, and motor current.

Fatigue Analysis of a Multi-Step Driveshaft September 2025 – October 2025

- Modeled combined bending and torsional stresses along a stepped driveshaft to identify critical fatigue locations under steady-state operations.
- Developed fatigue life predictions using both Morrow and Smith–Watson–Topper corrections, selecting models based on material behavior and mean stresses.
- Formulated a nonlinear constrained optimization problem to minimize shaft mass by iterating on section diameters and fillet radii while enforcing fatigue life requirements.
- Applied design constraints such as minimum step changes, notch sensitivity factors, and conservative load factors to ensure a feasible solution.

Topology Optimization of a Complex Truss System March 2025 – May 2025

- Developed a custom 2D finite element solver in MATLAB to model the stress and strain in a nonlinear truss system under static loading.
- Implemented multidimensional Newton–Raphson to numerically solve the nonlinear system.
- Coupled the numerical solver with a genetic algorithm to perform topology optimization, minimizing material usage subject to displacement constraints and continuity conditions.
- Verified solver accuracy by benchmarking reaction forces and deflections against analytical solutions and linear finite element models across increasing complexity.

Model Predictive Control of a Mass-Spring-Damper System January 2025 – February 2025

- Implemented nonlinear and pseudo-linear cart models in MATLAB and compared their responses to the same control and disturbance inputs.
- Formulated an MPC quadratic program with system-dynamics constraints and +/-5 V motor-input limits, using a course-provided solver.
- Implemented the controller in the MATLAB/Simulink laboratory framework and compared simulated and experimental reference tracking under an applied disturbance.